

Builder Basics · Diamond Structure (Final Project) — English Worksheet (Beginner)

Topic: Diamond Structure (Final Project) · **Course:** Builder Basics · **Level:** Beginner · **Time:** about 45 minutes

A diamond is two pyramids: the **bottom half grows** ($1 \rightarrow 3 \rightarrow 5$) and the **top half shrinks** ($3 \rightarrow 1$).

1 · Predict 🧠

Read the steps. Circle your answer.

```
# Bottom half
set size to 1
while size <= 5:
    [build layer]
    move up by 1
    set size to size + 2
```

The bottom half grows or shrinks? Circle one: grows · shrinks

Layer sizes in order? Circle one: $1 \cdot 3 \cdot 5 \rightarrow 5 \cdot 3 \cdot 1$

```
# Top half
set size to 3
while size > 0:
    [build layer]
    move up by 1
    set size to size - 2
```

The top half grows or shrinks? Circle one: grows · shrinks

Why start at 3, not 5? Circle one: the size-5 layer is already built · 5 is too big to build · the agent cannot count to 5

2 · Find the Difference 🐉

Clean steps first, then a buggy version. Circle the bug.

Pair A — The top half should start at **3**, so the widest layer is not built twice.

```
# clean
set size to 3
while size > 0:
    [build layer]
    move up by 1
    set size to size - 2
```

```
# buggy
set size to 5
while size > 0:
    [build layer]
    move up by 1
    set size to size - 2
```

What is wrong? Which layer is built twice? Circle one: the size-5 layer · the size-3 layer · the size-1 layer

Pair B — The bottom half must **move up** after each layer.

```
# clean
while size <= 5:
    [build layer]
    move up by 1
    set size to size + 2
```

```
# buggy
while size <= 5:
    [build layer]
    set size to size + 2
```

What is wrong? Circle one: it never moves up · it moves up too far · it turns right

3 · Fill the Gap

Here is the diamond skeleton. Fill the missing number.

```
# Bottom half (grows 1 → 3 → 5)
set size to 1
while size <= 5:
    [build layer]
    move up by 1
    set size to size + 2

# Top half (shrinks 3 → 1)
set size to ____
while size > 0:
    [build layer]
    move up by 1
    set size to size - 2
```

Missing number? Circle one: 3 · 5 · 1

4 · Show Your Work 📷 🎥

Build your **final project**: a diamond with a bottom half that grows **1** → **3** → **5** and a top half that shrinks **3** → **1**.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Say these out loud, filling in the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.